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Xin Liu

Senior Applied Scientist, Amazon Search/A9

Job Duties: Research and Development on E-commerce Foundation Models and Agentic Capabilities

Education

Aug. 2018 - July 2023 — **Ph.D. Degree** at Department of Computer Science and Engineering, Hong Kong University of Science and Technology, Hong Kong SAR, supervised by Prof. Yangqiu Song

Aug. 2014 - June 2018 — **B.E. Degree** at Department of Computer Science, Sun Yat-sen University, China, supervised by Prof. Rong Pan

Experience

Fulltime

Aug. 2023 - Present — **Applied Scientist – Senior Applied Scientist (Foundation Model)** [Manager: Dr. Qingyu Yin] Store Foundational AI, Amazon Search, Palo Alto.

- **Foundation Model Development:** Architected the complete data recipes and training strategies for (dense & sparse) foundational models from scaling laws to the trillion parameter model, covering the full lifecycle from pre-training from scratch to continual pre-training and annealing. Spearheaded alignment and fine-tuning processes during post-training to refine instruction following, instruction decomposition, conversational flow, reasoning, and shopping missions, from the view of input structures, capability taxonomy, and data efficiency.
- **Agentic Reinforcement Learning:** Lead RLVR development by collecting and augmenting millions of data seeds and research on data difficulty distribution and learning efficiency. Designed an on-policy query augmentation pipeline and difficulty-based samplers to scale training efficiency. Improved agent capabilities with logical query diversification and multi-tool orchestration rewards.
- **Generative AI Application & Infrastructure:** Played a pivotal role in launching Amazon's first generative AI application, **Rufus**. Designed and developed the online prompting and routing platform and an end-to-end debugging environment to ensure production readiness and quality. Babysitted MoE models with thousands of GPU training from stability and throughput.
- **Performance Optimization:** Enhanced model stability and performance in pre-training and post-training by data curriculum, best-fit-packing, instruction composition and decomposition, prompt structure augmentation. Optimized the generation and evaluation efficiency by profiling and caching in inference.

Internship

June 2022 - Dec. 2022 — **Research Intern (Search)** [Mentor: Dr. Zheng Li], Search Mission Understanding, Amazon Search, Palo Alto.

Jan. 2018 - July 2018 — **Research Assistant (Text Mining)** [Mentor: Prof. Yangqiu Song], Knowledge Computation Group, Hong Kong University of Science and Technology, Hong Kong SAR.

June 2017 - Dec. 2017 — **Research Intern (Cloud Graph Database)** [Mentor: Dr. Liang Jeff Chen], Systems Research Group, Microsoft Research Asia, China.

Academic

Journal Reviewer: DMKD'23 TKDE'22,'23

Experience (continued)

- **Program Committee Member:** CIKM'19,'21,'22,'23 IJCAI'21,'22,'23 AAAI'21,'22,'23,'24 NAACL'22 ACL ARR'22,'23,'24,'25 ICML'22,'24 SIGKDD'19,'20,'21,'22,'23,'24,'25 NeurIPS'22,'23,'24,'25 ICLR'24,'25 COLM'24,'25

Awards

2022	HKUST RedBird Academic Excellence Award
2020	HKUST SENG Academic Award
	HKUST CSE Professor Samuel Chanson Best Teaching Assistant Award
2018	SYSU Outstanding Graduate
	SYSU Outstanding Graduate Thesis
	The First Prize, Guangdong Collegiate Programming Contest
2017	Meritorious Winner, Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling
2015, 2016, 2017	National Scholarship ×3

Publications

* means equal contribution.

Journal Articles

1. Zhongwei Wan, **Xin Liu**, Benyou Wang, Jiezhong Qiu, Boyu Li, Ting Guo, Guangyong Chen and Yang Wang. 'Spatio-Temporal Contrastive Learning Enhanced GNNs for Session-based Recommendation', 2023.
2. Hongming Zhang*, **Xin Liu***, Haojie Pan*, Haowen Ke, Jiefu Ou, Tianqing Fang and Yangqiu Song. 'ASER: Towards Large-scale Commonsense Knowledge Acquisition via Higher-order Selectional Preference over Eventualities'. *Artificial Intelligence (AIJ)*, 2022.

Conference Publications

1. Chi Han, **Xin Liu**, Haodong Wang, Shiyang Li, Jingfeng Yang, Haoming Jiang, Zhengyang Wang et al. 'Can Language Models Follow Multiple Turns of Entangled Instructions?' In *Findings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2025.
2. Fenglin Liu, Zheng Li, Hongjian Zhou, Qingyu Yin, Jingfeng Yang, **Xin Liu**, Zhengyang Wang, Xianfeng Tang, Shiyang Li et al. 'DrAgent: Empowering Large Language Models as Medical Agents for Multi-hop Medical Reasoning'. In *Findings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2025.
3. Jiayang Cheng, Qianqian Zhuang, Haoran Li, Chunkit Chan, **Xin Liu**, Lin Qiu and Yangqiu Song. 'InteGround: On the Evaluation of Verification and Retrieval Planning in Integrative Grounding'. In *Findings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2025.
4. Weiqi Wang, Tianqing Fang, Haochen Shi, Baixuan Xu, Wenxuan Ding, Liyu Zhang, Wei Fan et al. 'On the Role of Entity and Event Level Conceptualization in Generalizable Reasoning: A Survey of Tasks, Methods, Applications, and Future Directions'. In *Findings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2025.
5. Fengran Mo, Yifan Gao, Chuan Meng, **Xin Liu**, Zhuofeng Wu, Kelong Mao, Zhengyang Wang, Pei Chen, Zheng Li, Xian Li et al. 'UniConv: Unifying Retrieval and Response Generation for Large Language Models in Conversations'. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*. 2025.
6. Weiqi Wang, Limeng Cui, **Xin Liu**, Sreyashi Nag, Wenju Xu, Chen Luo, Sheikh Muhammad Sarwar, Yang Li, Hansu Gu, Hui Liu et al. 'EcomScriptBench: A multi-task benchmark for e-commerce script planning via step-wise intention-driven product association'. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*. 2025.
7. Yuchen Zhuang, Jingfeng Yang, Haoming Jiang, **Xin Liu**, Kewei Cheng, Sanket Lokegaonkar, Yifan Gao, Qing Ping, Tianyi Liu, Binxuan Huang et al. 'Hephaestus: Improving Fundamental Agent Capabilities of Large Language Models through Continual Pre-Training'. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*. 2025.
8. Zhihan Zhang, Shiyang Li, Zixuan Zhang, **Xin Liu**, Haoming Jiang, Xianfeng Tang, Yifan Gao, Zheng Li, Haodong Wang, Zhaoxuan Tan et al. 'IHEval: Evaluating language models on following the instruction hierarchy'. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*. 2025.

9 Siqi Guo, Ilgee Hong, Vicente Balmaseda, Changlong Yu, Liang Qiu, **Xin Liu**, Haoming Jiang, Tuo Zhao and Tianbao Yang. 'Discriminative Finetuning of Generative Large Language Models without Reward Models and Human Preference Data'. In *Proceedings of the International Conference on Machine Learning (ICML)*. 2025.

10 Yilun Jin, Zheng Li, Chenwei Zhang, Tianyu Cao, Yifan Gao, Pratik Jayarao, Mao Li, **Xin Liu**, Ritesh Sarkhel, Xianfeng Tang et al. 'Shopping mmlu: A massive multi-task online shopping benchmark for large language models'. In *Conference on Neural Information Processing Systems, Datasets and Benchmarks Track (NeurIPS)*. 2024.

11 Zhaowei Wang, Haochen Shi, Weiqi Wang, Tianqing Fang, Hongming Zhang, Sehyun Choi, **Xin Liu** and Yangqiu Song. 'AbsPyramid: Benchmarking the Abstraction Ability of Language Models with a Unified Entailment Graph'. In *Findings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*. 2024.

12 Chunkit Chan, Cheng Jiayang, Yauwai Yim, Zheye Deng, Wei Fan, Haoran Li, **Xin Liu**, Hongming Zhang, Weiqi Wang and Yangqiu Song. 'NegotiationToM: A Benchmark for Stress-testing Machine Theory of Mind on Negotiation Surrounding'. In *Findings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2024.

13 Wenxuan Ding, Weiqi Wang, Sze Kwok, Minghao Liu, Tianqing Fang, Jiaxin Bai, **Xin Liu**, Changlong Yu, Zheng Li, Chen Luo et al. 'IntentionQA: A Benchmark for Evaluating Purchase Intention Comprehension Abilities of Language Models in E-commerce'. In *Findings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2024.

14 Chunkit Chan, Jiayang Cheng, Weiqi Wang, Yuxin Jiang, Tianqing Fang, **Xin Liu** and Yangqiu Song. 'Exploring the potential of ChatGPT on sentence level relations: A focus on temporal, causal, and discourse relations'. In *Findings of the European Chapter of the Association for Computational Linguistics (EACL)*. 2024.

15 Chunkit Chan, Jiayang Cheng, Weiqi Wang, Yuxin Jiang, Tianqing Fang, **Xin Liu** and Yangqiu Song. 'ChatGPT Evaluation on Sentence Level Relations: A Focus on Temporal, Causal, and Discourse Relations'. In *Findings of the European Chapter of the Association for Computational Linguistics (EACL)*. 2024.

16 Zizheng Lin, Chunkit Chan, Yangqiu Song and **Xin Liu**. 'Constrained reasoning chains for enhancing theory-of-mind in large language models'. In *Pacific Rim International Conference on Artificial Intelligence (PRICAI)*. Springer Nature Singapore Singapore, 2024.

17 Changlong Yu*, **Xin Liu***, Jefferson Maia, Tianyu Cao, Yang Li, Yifan Gao, Yangqiu Song et al. 'COSMO: A large-scale e-commerce common sense knowledge generation and serving system at Amazon'. In *Proceedings of the ACM SIGMOD International Conference on Management of Data (SIGMOD)*. 2024.

18 Haoran Li, Dadi Guo, Donghao Li, Wei Fan, Qi Hu, **Xin Liu**, Chunkit Chan, Duanyi Yao and Yangqiu Song. 'PrivLM-Bench: A Multi-level Privacy Evaluation Benchmark for Language Models'. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*. 2024.

19 Jiaxin Bai, Yicheng Wang, Tianshi Zheng, Yue Guo, **Xin Liu** and Yangqiu Song. 'Advancing Abductive Reasoning in Knowledge Graphs through Complex Logical Hypothesis Generation'. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*. 2024.

20 Weiqi Wang, Tianqing Fang, Chunyang Li, Haochen Shi, Wenxuan Ding, Baixuan Xu, Zhaowei Wang et al. 'CANDLE: Iterative Conceptualization and Instantiation Distillation from Large Language Models for Commonsense Reasoning'. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*. 2024.

21 Zhaowei Wang, Wei Fan, Qing Zong, Hongming Zhang, Sehyun Choi, Tianqing Fang, **Xin Liu**, Yangqiu Song, Ginny Y. Wong and Simon See. 'AbsInstruct: Eliciting Abstraction Ability from LLMs through Explanation Tuning with Plausibility Estimation'. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*. 2024.

22 Baixuan Xu, Weiqi Wang, Haochen Shi, Wenxuan Ding, Huihao Jing, Tianqing Fang, Jiaxin Bai, **Xin Liu**, Changlong Yu, Zheng Li et al. 'MIND: Multimodal Shopping Intention Distillation from Large Vision-language Models for E-commerce Purchase Understanding'. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2024.

23 Chunkit Chan, Jiayang Cheng, **Xin Liu**, Yauwai Yim, Yuxin Jiang, Zheye Deng, Haoran Li, Yangqiu Song, Ginny Wong and Simon See. 'Audience Persona Knowledge-Aligned Prompt Tuning Method for Online Debate'. In *Proceedings of the European Conference on Artificial Intelligence (ECAI)*. 2024.

24 Jiayang Cheng, Lin Qiu, Chunkit Chan, **Xin Liu**, Yangqiu Song and Zheng Zhang. 'EventGround: Narrative Reasoning by Grounding to Eventuality-centric Knowledge Graphs'. In *Proceedings of the International Conference on Computational Linguistics (COLING)*. 2024.

25 Hejie Cui, Xinyu Fang, Zihan Zhang, Ran Xu, Xuan Kan, **Xin Liu**, Yue Yu, Manling Li, Yangqiu Song and Carl Yang. 'Open Visual Knowledge Extraction via Relation-Oriented Multimodality Model Prompting'. In *Conference on Neural Information Processing Systems (NeurIPS)*. 2023.

26 Jiaxin Bai, **Xin Liu**, Weiqi Wang, Chen Luo and Yangqiu Song. 'Complex Eventuality Reasoning with Implicit Logical Constraints'. In *Conference on Neural Information Processing Systems (NeurIPS)*. 2023.

27 **Xin Liu**, Zheng Li, Yifan Gao, Jingfeng Yang, Tianyu Cao, Zhengyang Wang, Bing Yin and Yangqiu Song. 'Enhancing User Intent Capture in Session-Based Recommendation with Attribute Patterns'. In *Conference on Neural Information Processing Systems (NeurIPS)*. 2023.

28 Zheye Deng, Weiqi Wang, Zhaowei Wang, **Xin Liu** and Yangqiu Song. 'Gold: A Global and Local-aware Denoising Framework for Commonsense Knowledge Graph Noise Detection'. In *Conference on Neural Information Processing Systems (NeurIPS)*. 2023.

29 Changlong Yu, Weiqi Wang, **Xin Liu**, Jiaxin Bai, Yangqiu Song, Zheng Li, Yifan Gao, Tianyu Cao and Bing Yin. 'FolkScope: Intention Knowledge Graph Construction for Discovering E-commerce Commonsense'. In *Findings of the Annual Meeting of the Association for Computational Linguistics (ACL)*. 2023.

30 Chunkit Chan*, **Xin Liu***, Jiayang Cheng, Zihan Li, Yangqiu Song, Ginny Y. Wong and Simon See. 'DiscoPrompt: Path Prediction Prompt Tuning for Implicit Discourse Relation Recognition'. In *Findings of the Annual Meeting of the Association for Computational Linguistics (ACL)*. 2023.

31 Haochen Shi*, Weiqi Wang*, Tianqing Fang, Baixuan Xu, Wenxuan Ding, **Xin Liu** and Yangqiu Song. 'QaDynamics: Training Dynamics-Driven Synthetic QA Diagnostic for Zero-Shot Commonsense Question Answering'. In *Findings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2023.

32 Weiqi Wang*, Tianqing Fang*, Wenxuan Ding, Baixuan Xu, **Xin Liu**, Yangqiu Song and Antoine Bosselut. 'CAR: Conceptualization-Augmented Reasoner for Zero-Shot Commonsense Question Answering'. In *Findings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2023.

33 Chunkit Chan, **Xin Liu**, Tsz Ho Chan, Jiayang Cheng, Yangqiu Song, Ginny Y. Wong and Simon See. 'Self-Consistent Narrative Prompts on Abductive Natural Language Inference'. In *Proceedings of the Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (AACL)*. 2023.

34 **Xin Liu** and Yangqiu Song. 'Graph Convolutional Networks with Dual Message Passing for Subgraph Isomorphism Counting and Matching'. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*. 2022.

35 Huiru Xiao, **Xin Liu**, Yangqiu Song, Ginny Y. Wong and Simon See. 'Complex Hyperbolic Knowledge Graph Embeddings with Fast Fourier Transform'. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2022.

36 **Xin Liu**, Jiayang Cheng, Yangqiu Song and Xin Jiang. 'Boosting Graph Structure Learning with Dummy Nodes'. In *Proceedings of the International Conference on Machine Learning (ICML)*. 2022.

37 **Xin Liu**, Jiefu Ou, Yangqiu Song and Xin Jiang. 'Exploring Discourse Structures for Argument Impact Classification'. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*. 2021.

38 **Xin Liu**, Haojie Pan, Mutian He, Yangqiu Song, Xin Jiang and Lifeng Shang. 'Neural Subgraph Isomorphism Counting'. In *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. 2020.

39 **Xin Liu**, Jiefu Ou, Yangqiu Song and Xin Jiang. 'On the Importance of Word and Sentence Representation Learning in Implicit Discourse Relation Classification'. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*. 2020.

40 Hongming Zhang*, **Xin Liu***, Haojie Pan*, Yangqiu Song and Cane Wing-Ki Leung. 'ASER: A Large-scale Eventuality Knowledge Graph'. In *Proceedings of The Web Conference (WWW)*. 2020.

41 Jie Huang, **Xin Liu** and Yangqiu Song. 'Hyper-Path-Based Representation Learning for Hyper-Networks'. In *Proceedings of the ACM International Conference on Information and Knowledge Management (CIKM)*. 2019.

42 Yan Liang, **Xin Liu**, Jianwen Zhang and Yangqiu Song. 'Relation Discovery with Out-of-Relation Knowledge Base as Supervision'. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*. 2019.

43 Ziqian Zeng, Wenxuan Zhou, **Xin Liu** and Yangqiu Song. 'A Variational Approach to Weakly Supervised Document-Level Multi-Aspect Sentiment Classification'. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*. 2019.

44 Huiru Xiao, **Xin Liu** and Yangqiu Song. 'Efficient Path Prediction for Semi-Supervised and Weakly Supervised Hierarchical Text Classification'. In *Proceedings of The Web Conference (WWW)*. 2019.

45 Ziqian Zeng*, **Xin Liu*** and Yangqiu Song. 'Biased Random Walk based Social Regularization for Word Embeddings'. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*. 2018.